

Bryce Grant | EE PhD Candidate

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EDUCATION & HONORS

Case Western Reserve University

Cleveland, OH

2nd Year PhD Student in Electrical Engineering (Robotics, Geometric Deep Learning)

Expected: May 2028

Graduate Advisor: [Dr. Peng \(Edward\) Wang](#)

Graduate Honors: NSF Graduate Research Fellowship Program (NSF GRFP) Recipient, Advanced to Candidacy

University of Kentucky – Magna Cum Laude

Lexington, KY

Dual B.S. in Computer Engineering and Electrical Engineering (CS, Math Minors)

Graduated: May 2024

Undergraduate Honors: Lexmark International Scholarship, Dallas and Betty Wade Scholarship, F. '19 – F '23 Dean's List

PUBLICATIONS

IROS 2025

Hangzhou, China

B. Grant and P. Wang (2025) Quaternion Approximate Networks for Enhanced Image Classification and Oriented Object Detection.

IEEE/RSJ International Conference on Intelligent Robots and Systems, Oral. [Preprint](#)

RELEVANT EXPERIENCE

Mercor

Remote

Part-Time Software Tooling Engineer

Jun. 2025 – Sept. 2025

- Built automated data pipelines integrating Snowflake, Airtable, and Insightful APIs to track RLHF task completion and expert productivity metrics across 1000+ reviewers
- Developed HEX dashboards with real-time budget tracking, margin analysis, and productivity metrics for multi-million dollar training projects
- Contributed as domain expert for robotics/CV tasks, writing evaluation rubrics for edge cases in embodied AI training

Case Western Reserve University

Cleveland, OH

Graduate Research Assistant

Aug. 2024 – Present

- Quaternion Approximate Networks:** Developed a quaternion-inspired deep learning framework for rotation-aware classification and detection. Implemented Hamilton product approximations with custom CUDA kernels, reducing convolutional complexity 4×. Introduced Independent Quaternion Batch Normalization (IQBN) and quaternion attention mechanisms for stable training. Achieved 95.1%, 76.8%, and 74.28% accuracy on CIFAR10/100 and ImageNet respectively with 74% fewer parameters than real-valued networks while achieving SOTA accuracy for quaternion networks on MS COCO, and DOTA v1
- Sequential Planning via Anchored Robotic Keypoints (SPARK):** Designed an interpretable planning framework representing tasks as symbolic “scores” of keypoint-anchored actions. Built a **multi-stage perception pipeline** (SAM, DINOv2, CLIP) for keypoint detection and integrated impedance control for compliant execution. The sequencer uses **fallback logic** (re-grasp, retract-retry) to recover from errors, enabling faster authoring of new robotic assembly tasks and improved robustness in long-horizon manipulation
- Investigating optimization 48 quaternion multiplication representations for improved geometric deep learning

HP

Vancouver, WA

PHD ML Intern

May 2024 – Aug. 2024

- Deployed a hierarchical anomaly detection pipeline on AWS EC2** for enterprise **multi-service cloud spend analysis**, combining LSTM forecasting with STL decomposition to flag anomalies across accounts
- Engineered a comparative RAG framework: benchmarked semantic vs. recursive chunking, BM25 vs. FAISS retrieval, and LLM query expansion strategies, with evaluation via coherence and NDCG metrics
- Built a conversational printer setup agent with Mistral-7B: implemented dynamic reranking and multi-stage error recovery dialogue management for robust troubleshooting assistance
- Developed semantic search over 10K+ customer calls using BERT embeddings and topic modeling for R&D insights

University of Kentucky

Lexington, KY

AI for Smart Manufacturing Lab Research Assistant

Aug. 2023 – May. 2024

- Developed CNN-Transformer hybrid for 6-DOF pose estimation from RGB-D inputs for robotic arm manipulation achieving 0.92 mAP50-95 on a custom assembly environment dataset and .72 mAP50-95 on DOTA v1

Honeywell

Atlanta, GA

<i>Embedded Software Engineer Intern</i>	May. 2023 – Aug. 2023
- Developed bare-metal firmware API for life safety microcontroller (TI MSP430), supporting a product line with 100k+ unit volume - Validated dual-CPU communication protocols using oscilloscope debugging and hardware-in-loop testing	
Xerox Lexmark	Lexington, KY
<i>Electrical Engineering Intern</i>	May 2022 – May 2023
Designed software-controlled test equipment to optimize print quality and fixture reliability for the Laser Scanning Unit	
ShopStock LLC	Louisville, KY
<i>Co-Founder and Embedded Systems Developer</i>	Sept. 2020 – Sept. 2022
Built hardware bridge using ESP-32 to retrofit legacy POS systems with real-time inventory tracking via custom firmware	

SELECTED PROJECTS

Causal PointNet
Refines 6D pose estimates using causal interventions and backdoor adjustments based on structural causal models while improving robustness to viewpoint ambiguity and symmetry
Probabilistic Digital Twin
Developed a ROS2-integrated Dynamic Bayesian Network for real-time fault detection in Universal Robots using Unscented Kalman Filtering to track friction, damping, and wear parameters
Just a Rather Very Intelligent (System (J.A.R.V.I.S.))
Led navigation subsystem for autonomous telepresence robot, implementing 3D SLAM with servo-mounted LiDAR, achieving real-time mapping and dynamic obstacle avoidance in dynamic environments

SKILLS & CERTIFICATIONS

Programming Languages: C/C++, CUDA, MATLAB, Python, SQL
ML & Robotics: FAISS, HuggingFace, MuJoCo, OpenCV, Open3D, Pybullet, Pytorch, ROS2, TensorFlow, WandB
Cloud & DevOps: AWS, Docker, FAISS, Linux, Snowflake
Engineering Software: Ansys Workbench, Autodesk EAGLE, SolidWorks, Wireshark
Domains & Methodologies: Causal Inference, Computer Vision, Geometric Deep Learning, NLP, Robotics, Sequential Modeling
Certifications: LEAN Systems Certification, CITI Responsible Conduct of Research
Languages: English (Native), French (Working Proficiency), Portuguese (Intermediate), Russian (Intermediate)

PROFESSIONAL & LEADERSHIP AFFILIATION

Mathworks	Cleveland, OH
<i>MATLAB Student Ambassador</i>	Oct. 2024 – Present
Organize workshops, ML hackathons, and campus outreach initiatives on MATLAB applications in research	
National Society of Black Engineers	Lexington, KY
<i>Region III Finance Chair</i>	May 2023 – May. 2024
- Managed a network of 250+ companies and organizations - Orchestrated a regional career fair with 50+ participating companies - Generated over \$250,000 in revenue within the first six months	

SELECTED COURSEWORK

AI for DB, Causality & Inference, Computer Architecture, Convex Optimization, Data Mining, Machine Learning & Process Automation, Probabilistic Graph Models, Robotics, Stochastic Modeling & Time Series
